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LaTeX:Symbols

LaTeX

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This article provides a short list of commonly used LaTeX symbols.

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Finding Other Symbols

Here are some external resources for finding less commonly used symbols:

- Detexify (<http://detexify.kirelabs.org/classify.html>) is an online application which allows you to draw the symbol you'd like and shows you the $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ code for it!
- MathJax (what allows us to use $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ on the web, (technically an AJAX library simulating it.)) maintains a list of supported commands (<http://docs.mathjax.org/en/latest/tex.html#supported-latex-commands>).
- The Comprehensive LaTeX Symbol List (<http://mirrors.ctan.org/info/symbols/comprehensive/symbols-a4.pdf>).
- Comprehensive List of Mathematical Symbols (<https://mathvault.ca/wp-content/uploads/Comprehensive-List-of-Mathematical-Symbols.pdf>).

Operators

Symbol	Command	Symbol	Command	Symbol	Command
$\pm$	<code>\pm</code>	$\mp$	<code>\mp</code>	$\times$	<code>\times</code>
$\div$	<code>\div</code>	$\cdot$	<code>\cdot</code>	$*$	<code>\ast</code>

★	<code>\star</code>	†	<code>\dagger</code>	‡	<code>\ddagger</code>
∏	<code>\amalg</code>	∩	<code>\cap</code>	∪	<code>\cup</code>
⊕	<code>\uplus</code>	⊔	<code>\sqcap</code>	⊔	<code>\sqcup</code>
∨	<code>\vee</code>	∧	<code>\wedge</code>	⊕	<code>\oplus</code>
⊖	<code>\ominus</code>	⊗	<code>\otimes</code>	○	<code>\circ</code>
•	<code>\bullet</code>	◇	<code>\diamond</code>	◁	<code>\lhd</code>
▷	<code>\rhd</code>	◁	<code>\unlhd</code>	▷	<code>\unrhd</code>
⊘	<code>\oslash</code>	⊙	<code>\odot</code>	◯	<code>\bigcirc</code>
◁	<code>\triangleleft</code>	◊	<code>\Diamond</code>	△	<code>\bigtriangleup</code>
▽	<code>\bigtriangledown</code>	□	<code>\Box</code>	▷	<code>\triangleright</code>
\	<code>\setminus</code>	}	<code>\wr</code>	$\sqrt{x}$	<code>\sqrt{x}</code>
x°	<code>x^{\circ}</code>	▽	<code>\triangledown</code>	$\sqrt[n]{x}$	<code>\sqrt[n]{x}</code>
a^x	<code>a^x</code>	a^{xyz}	<code>a^{xyz}</code>	a_x	<code>a_x</code>

Relations

Symbol	Command	Symbol	Command	Symbol	Command
$\leq$	<code>\le</code>	$\geq$	<code>\ge</code>	$\neq$	<code>\neq</code>
$\sim$	<code>\sim</code>	$\ll$	<code>\ll</code>	$\gg$	<code>\gg</code>
$\doteq$	<code>\doteq</code>	$\simeq$	<code>\simeq</code>	$\subset$	<code>\subset</code>
$\supset$	<code>\supset</code>	$\approx$	<code>\approx</code>	$\asymp$	<code>\asymp</code>
$\subseteq$	<code>\subseteq</code>	$\supseteq$	<code>\supseteq</code>	$\cong$	<code>\cong</code>
$\smile$	<code>\smile</code>	$\sqsubset$	<code>\sqsubset</code>	$\sqsupset$	<code>\sqsupset</code>
$\equiv$	<code>\equiv</code>	$\frown$	<code>\frown</code>	$\sqsubseteq$	<code>\sqsubseteq</code>
$\sqsupseteq$	<code>\sqsupseteq</code>	$\propto$	<code>\propto</code>	$\bowtie$	<code>\bowtie</code>
$\in$	<code>\in</code>	$\ni$	<code>\ni</code>	$\prec$	<code>\prec</code>
$\succ$	<code>\succ</code>	$\vdash$	<code>\vdash</code>	$\dashv$	<code>\dashv</code>
$\preceq$	<code>\preceq</code>	$\succeq$	<code>\succeq</code>	$\models$	<code>\models</code>
$\perp$	<code>\perp</code>	$\parallel$	<code>\parallel</code>		
$\mid$	<code>\mid</code>	$\bumpeq$	<code>\bumpeq</code>		

Negations of many of these relations can be formed by just putting `\not` before the symbol, or by slipping an "n" between the `\` and the word. Here are a couple examples, plus many other negations; it works for many of the many others as well.

Symbol	Command	Symbol	Command	Symbol	Command
$\nmid$	<code>\nmid</code>	$\nleq$	<code>\nleq</code>	$\ngeq$	<code>\ngeq</code>
$\nsim$	<code>\nsim</code>	$\ncong$	<code>\ncong</code>	$\nparallel$	<code>\nparallel</code>
$\not<$	<code>\not<</code>	$\not>$	<code>\not></code>	$\not=$ or $\not\neq$ or $\not\neq$	
$\not\leq$	<code>\not\le</code>	$\not\geq$	<code>\not\ge</code>	$\not\sim$	<code>\not\sim</code>
$\not\approx$	<code>\not\approx</code>	$\not\cong$	<code>\not\cong</code>	$\not\equiv$	<code>\not\equiv</code>
$\not\parallel$	<code>\not\parallel</code>	$\nless$	<code>\nless</code>	$\ngtr$	<code>\ngtr</code>
$\lneq$	<code>\lneq</code>	$\gneq$	<code>\gneq</code>	$\lnsim$	<code>\lnsim</code>
$\lneqq$	<code>\lneqq</code>	$\gneqq$	<code>\gneqq</code>		

To use other relations not listed here, such as $=$, $>$, and $<$, in LaTeX, you must use the symbols on your keyboard, as they are not available in **L^AT_EX**.

Greek Letters

Lowercase Letters

Symbol	Command	Symbol	Command	Symbol	Command	Symbol	Command
α	<code>\alpha</code>	β	<code>\beta</code>	γ	<code>\gamma</code>	δ	<code>\delta</code>

ϵ	<code>\epsilon</code>	ε	<code>\varepsilon</code>	ζ	<code>\zeta</code>	η	<code>\eta</code>
θ	<code>\theta</code>	ϑ	<code>\vartheta</code>	ι	<code>\iota</code>	κ	<code>\kappa</code>
λ	<code>\lambda</code>	μ	<code>\mu</code>	ν	<code>\nu</code>	ξ	<code>\xi</code>
π	<code>\pi</code>	ϖ	<code>\varpi</code>	ρ	<code>\rho</code>	ϱ	<code>\varrho</code>
σ	<code>\sigma</code>	ς	<code>\varsigma</code>	τ	<code>\tau</code>	υ	<code>\upsilon</code>
ϕ	<code>\phi</code>	φ	<code>\varphi</code>	χ	<code>\chi</code>	ψ	<code>\psi</code>
ω	<code>\omega</code>						

Capital Letters

Symbol Command Symbol Command Symbol Command Symbol Command

Γ	<code>\Gamma</code>	Δ	<code>\Delta</code>	Θ	<code>\Theta</code>	Λ	<code>\Lambda</code>
Ξ	<code>\Xi</code>	Π	<code>\Pi</code>	Σ	<code>\Sigma</code>	Υ	<code>\Upsilon</code>
Φ	<code>\Phi</code>	Ψ	<code>\Psi</code>	Ω	<code>\Omega</code>		

Arrows

Symbol	Command	Symbol	Command
$\leftarrow$	<code>\gets</code>	$\rightarrow$	<code>\to</code>
$\leftarrow$	<code>\leftarrow</code>	$\Leftarrow$	<code>\Leftarrow</code>
$\rightarrow$	<code>\rightarrow</code>	$\Rightarrow$	<code>\Rightarrow</code>
$\leftrightarrow$	<code>\leftrightarrow</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\mapsto$	<code>\mapsto</code>	$\hookrightarrow$	<code>\hookrightarrow</code>
$\leftharpoonup$	<code>\leftharpoonup</code>	$\leftharpoondown$	<code>\leftharpoondown</code>
$\rightrightarrows$	<code>\rightrightarrows</code>	$\longleftarrow$	<code>\longleftarrow</code>
$\Longleftarrow$	<code>\Longleftarrow</code>	$\longrightarrow$	<code>\longrightarrow</code>
$\Longrightarrow$	<code>\Longrightarrow</code>	$\longleftrightarrow$	<code>\longleftrightarrow</code>
$\Longleftrightarrow$	<code>\Longleftrightarrow</code>	$\longmapsto$	<code>\longmapsto</code>
$\hookrightarrow$	<code>\hookrightarrow</code>	$\rightharpoonup$	<code>\rightharpoonup</code>
$\rightharpoondown$	<code>\rightharpoondown</code>	$\leadsto$	<code>\leadsto</code>
$\uparrow$	<code>\uparrow</code>	$\Uparrow$	<code>\Uparrow</code>
$\downarrow$	<code>\downarrow</code>	$\Downarrow$	<code>\Downarrow</code>
$\updownarrow$	<code>\updownarrow</code>	$\Updownarrow$	<code>\Updownarrow</code>
$\nearrow$	<code>\nearrow</code>	$\searrow$	<code>\searrow</code>
$\swarrow$	<code>\swarrow</code>	$\nwarrow$	<code>\nwarrow</code>
$\overrightarrow{AB}$	<code>\overrightarrow{AB}</code>	$\overleftarrow{AB}$	<code>\overleftarrow{AB}</code>
$\overleftrightarrow{AB}$	<code>\overleftrightarrow{AB}</code>		

(For those of you who hate typing long strings of letters, `\iff` and `\implies` can be used in place of `\Longleftrightarrow` and `\Longrightarrow` respectively.)

Dots

Symbol Command Symbol Command

$\cdot$	<code>\cdot</code>	$\vdots$	<code>\vdots</code>
$\dots$	<code>\dots</code>	$\ddots$	<code>\ddots</code>
$\cdots$	<code>\cdots</code>	$\iddots$	<code>\iddots</code>

Accents

Symbol Command Symbol Command Symbol Command

$\hat{x}$	<code>\hat{x}</code>	$\check{x}$	<code>\check{x}</code>	$\dot{x}$	<code>\dot{x}</code>
$\breve{x}$	<code>\breve{x}</code>	$\acute{x}$	<code>\acute{x}</code>	$\ddot{x}$	<code>\ddot{x}</code>
$\grave{x}$	<code>\grave{x}</code>	$\tilde{x}$	<code>\tilde{x}</code>	$\mathring{x}$	<code>\mathring{x}</code>
$\bar{x}$	<code>\bar{x}</code>	$\vec{x}$	<code>\vec{x}</code>		

When applying accents to i and j, you can use `\imath` and `\jmath` to keep the dots from interfering with the accents:

Symbol	Command	Symbol	Command
$\vec{j}$	<code>\vec{\jmath}</code>	$\tilde{i}$	<code>\tilde{\imath}</code>

`\tilde` and `\hat` have wide versions that allow you to accent an expression:

Symbol	Command	Symbol	Command
$\widehat{7+x}$	<code>\widehat{7+x}</code>	$\widetilde{abc}$	<code>\widetilde{abc}</code>

Others

Symbol	Command	Symbol	Command	Symbol	Command
∞	<code>\infty</code>	$\triangle$	<code>\triangle</code>	$\angle$	<code>\angle</code>
$\aleph$	<code>\aleph</code>	$\hbar$	<code>\hbar</code>	$\imath$	<code>\imath</code>
$\jmath$	<code>\jmath</code>	ℓ	<code>\ell</code>	$\wp$	<code>\wp</code>
$\Re$	<code>\Re</code>	$\Im$	<code>\Im</code>	$\mho$	<code>\mho</code>
$\prime$	<code>\prime</code>	$\emptyset$	<code>\emptyset</code>	∇	<code>\nabla</code>
$\surd$	<code>\surd</code>	∂	<code>\partial</code>	$\top$	<code>\top</code>
$\bot$	<code>\bot</code>	$\vdash$	<code>\vdash</code>	$\dashv$	<code>\dashv</code>
$\forall$	<code>\forall</code>	$\exists$	<code>\exists</code>	$\neg$	<code>\neg</code>
$\flat$	<code>\flat</code>	$\natural$	<code>\natural</code>	$\sharp$	<code>\sharp</code>
$\backslash$	<code>\backslash</code>	$\Box$	<code>\Box</code>	$\Diamond$	<code>\Diamond</code>
$\clubsuit$	<code>\clubsuit</code>	$\diamondsuit$	<code>\diamondsuit</code>	$\heartsuit$	<code>\heartsuit</code>
$\spadesuit$	<code>\spadesuit</code>	$\Join$	<code>\Join</code>	$\blacksquare$	<code>\blacksquare</code>
$\diamondsuit$	<code>\diamondsuit</code>	$\copyright$	<code>\copyright</code>	$\underarc{XYZ}$	<code>\underarc{XYZ}</code>
$\heartsuit$	<code>\heartsuit</code>	$\overarc{ABC}$	<code>\overarc{ABC}</code>	$\cup$	<code>\cup</code>
$\S$	<code>\S</code>	$\P$	<code>\P</code>	$\Vdash$	<code>\Vdash</code>
$\pounds$	<code>\pounds</code>	$\in$	<code>\in</code>	$\Vdash$	<code>\Vdash</code>
$\bigstar$	<code>\bigstar</code>	$\implies$	<code>\implies</code>	$\LaTeX$	<code>\LaTeX</code>
$\square$	<code>\square</code>	$\text{\LaTeX}$	<code>\text{\LaTeX}</code>		
$\smiley$	<code>\smiley</code>				
$\mathbb{R}$	<code>\mathbb{R}</code> (represents all real numbers)				
$\checkmark$	<code>\checkmark</code>				
$\cancer$	<code>\cancer</code>				

Note: $\cancer$ and $\overarc{ABC}$ do not work in the classroom.

Command Symbols

Some symbols are used in commands, so they need to be treated in a special way.

Symbol	Command	Symbol	Command	Symbol	Command	Symbol	Command
$\$$	<code>\textdollar</code> or <code>\$</code>	$\&$	<code>\&</code>	$\%$	<code>\%</code>	$\#$	<code>\#</code>
$-$	<code>_</code>	$\{$	<code>\{</code>	$\}$	<code>\}</code>	$\backslash$	<code>\backslash</code>

(Warning: Using \$ for \$ will result in \$. This is a bug as far as we know. Depending on the version of **L^AT_EX** this is not always a problem.)

European Language Symbols

Symbol Command Symbol Command Symbol Command Symbol Command

œ	<code>\oe</code>	æ	<code>\ae</code>	ø	<code>\o</code>		
Œ	<code>\OE</code>	Æ	<code>\AE</code>	Å	<code>\AA</code>	Ø	<code>\O</code>
l	<code>\l</code>	ß	<code>\ss</code>	ı	<code>\i</code>		
Ł	<code>\L</code>	Š	<code>\SS</code>				

Bracketing Symbols

In mathematics, sometimes we need to enclose expressions in brackets, braces or parentheses. Some of these work just as you'd imagine in LaTeX; type (and) for parentheses, [and] for brackets, and | and | for absolute value. However, other symbols have special commands:

Symbol Command Symbol Command Symbol Command

{	<code>\{</code>	}	<code>\}</code>		<code>\ </code>
\	<code>\backslash</code>	⌊	<code>\lfloor</code>	⌋	<code>\rfloor</code>
⌈	<code>\lceil</code>	⌉	<code>\rceil</code>	⟨	<code>\langle</code>
⟩	<code>\rangle</code>				

You might notice that if you use any of these to typeset an expression that is vertically large, like

$$(\frac{a}{x})^2$$

the parentheses don't come out the right size:

$$\left(\frac{a}{x}\right)^2$$

If we put `\left` and `\right` before the relevant parentheses, we get a prettier expression:

$$\left(\frac{a}{x}\right)^2$$

gives

$$\left(\frac{a}{x}\right)^2$$

For systems of equations or piecewise functions, use the `cases` environment:

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ x & \text{if } x < 0 \end{cases}$$

which gives

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ x & \text{if } x < 0 \end{cases}$$

In addition to the `\left` and `\right` commands, when doing floor or ceiling functions with fractions, using

$$\left\lceil \frac{x}{y} \right\rceil$$

and $\left\lfloor \frac{x}{y} \right\rfloor$

gives both $\left\lceil \frac{x}{y} \right\rceil$ and $\left\lfloor \frac{x}{y} \right\rfloor$, respectively.

And, if you type this

`\underbrace{a_0+a_1+a_2+\cdots+a_n}_{x}`

Gives

$$\underbrace{a_0 + a_1 + a_2 + \cdots + a_n}_x$$

Or

`\overbrace{a_0+a_1+a_2+\cdots+a_n}^x`

Gives

$$\overbrace{a_0 + a_1 + a_2 + \cdots + a_n}^x$$

`\left` and `\right` can also be used to resize the following symbols:

Symbol	Command	Symbol	Command	Symbol	Command
$\uparrow$	<code>\uparrow</code>	$\downarrow$	<code>\downarrow</code>	$\Updownarrow$	<code>\updownarrow</code>
$\Uparrow$	<code>\Uparrow</code>	$\Downarrow$	<code>\Downarrow</code>	$\Updownarrow$	<code>\Updownarrow</code>

Multi-Size Symbols

Some symbols render differently in inline math mode and in display mode. Display mode occurs when you use `\[...]` or `$$...$$`, or environments like

...

,

...

. Read more in the commands section of the guide about how symbols which take arguments above and below the symbols, such as a summation symbol, behave in the two modes.

In each of the following, the two images show the symbol in display mode, then in inline mode.

Symbol	Command	Symbol	Command	Symbol	Command
Σ	<code>\sum</code>	$\int$	<code>\int</code>	$\oint$	<code>\oint</code>
$\prod$	<code>\prod</code>	$\coprod$	<code>\coprod</code>	$\bigcap$	<code>\bigcap</code>
$\bigcup$	<code>\bigcup</code>	$\bigsqcup$	<code>\bigsqcup</code>	$\bigvee$	<code>\bigvee</code>
$\bigwedge$	<code>\bigwedge</code>	$\bigodot$	<code>\bigodot</code>	$\bigotimes$	<code>\bigotimes</code>
$\bigoplus$	<code>\bigoplus</code>	$\biguplus$	<code>\biguplus</code>		

See More

- Next: Commands

Thanks for taking the time to scroll through and read all this! Here's a treat:

https://t3.ftcdn.net/jpg/01/65/99/48/360_F_165994815_BzUNaOtcF2jVwXY2cVD6jaiUH5UTSPV.jpg

Retrieved from "https://artofproblemsolving.com/wiki/index.php?title=LaTeX:Symbols&oldid=224495"